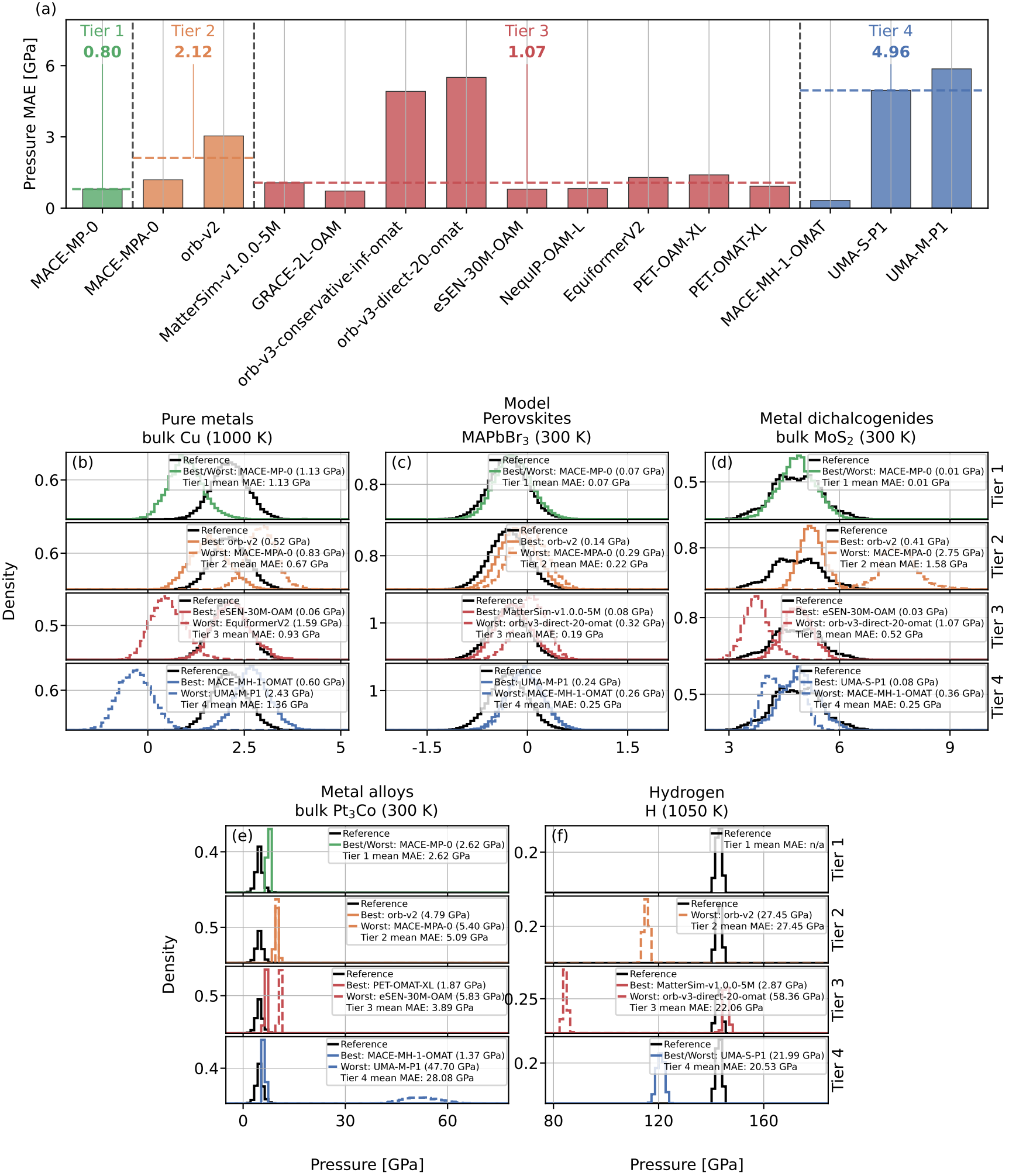
(a)
Tier 1
0.80
Tier 2
2.12
Tier 3
1.07
Tier 4
4.96
Pressure MAE [GPa]
MACE-MP-0
MACE-MPA-0
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1
UMA-M-P1
Pure metals
bulk Cu (1000 K)
Model
Perovskites
MAPbBr3 (300 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
(c)
(d)
Reference
Best/Worst: MACE-MP-0 (1.13 GPa)
Tier 1 mean MAE: 1.13 GPa
Best/Worst: MACE-MP-0 (0.07 GPa)
Tier 1 mean MAE: 0.07 GPa
Best/Worst: MACE-MP-0 (0.01 GPa)
Tier 1 mean MAE: 0.01 GPa
Best: orb-v2 (0.52 GPa)
Worst: MACE-MPA-0 (0.83 GPa)
Tier 2 mean MAE: 0.67 GPa
Best: orb-v2 (0.14 GPa)
Worst: MACE-MPA-0 (0.29 GPa)
Tier 2 mean MAE: 0.22 GPa
Best: orb-v2 (0.41 GPa)
Worst: MACE-MPA-0 (2.75 GPa)
Tier 2 mean MAE: 1.58 GPa
Best: eSEN-30M-OAM (0.06 GPa)
Worst: EquiformerV2 (1.59 GPa)
Tier 3 mean MAE: 0.93 GPa
Best: MatterSim-v1.0.0-5M (0.08 GPa)
Worst: orb-v3-direct-20-omat (0.32 GPa)
Tier 3 mean MAE: 0.19 GPa
Best: eSEN-30M-OAM (0.03 GPa)
Worst: orb-v3-direct-20-omat (1.07 GPa)
Tier 3 mean MAE: 0.52 GPa
Best: MACE-MH-1-OMAT (0.60 GPa)
Worst: UMA-M-P1 (2.43 GPa)
Tier 4 mean MAE: 1.36 GPa
Best: UMA-M-P1 (0.24 GPa)
Worst: MACE-MH-1-OMAT (0.26 GPa)
Tier 4 mean MAE: 0.25 GPa
Best: UMA-S-P1 (0.08 GPa)
Worst: MACE-MH-1-OMAT (0.36 GPa)
Tier 4 mean MAE: 0.25 GPa
Density
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk Pt3Co (300 K)
Hydrogen
H (1050 K)
(e)
(f)
Best/Worst: MACE-MP-0 (2.62 GPa)
Tier 1 mean MAE: 2.62 GPa
Tier 1 mean MAE: n/a
Best: orb-v2 (4.79 GPa)
Worst: MACE-MPA-0 (5.40 GPa)
Tier 2 mean MAE: 5.09 GPa
Worst: orb-v2 (27.45 GPa)
Tier 2 mean MAE: 27.45 GPa
Best: PET-OMAT-XL (1.87 GPa)
Worst: eSEN-30M-OAM (5.83 GPa)
Tier 3 mean MAE: 3.89 GPa
Best: MatterSim-v1.0.0-5M (2.87 GPa)
Worst: orb-v3-direct-20-omat (58.36 GPa)
Tier 3 mean MAE: 22.06 GPa
Best: MACE-MH-1-OMAT (1.37 GPa)
Worst: UMA-M-P1 (47.70 GPa)
Tier 4 mean MAE: 28.08 GPa
Best/Worst: UMA-S-P1 (21.99 GPa)
Tier 4 mean MAE: 20.53 GPa
Pressure [GPa]
Pressure [GPa]